Zoe Hackshaw

Curriculum Vitae

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EDUCATION

Ph.D. in Astronomy, The University of Texas at Austin
Advisor: Dr. Keith Hawkins
Expected June 2027

M.A. in Astronomy, The University of Texas at Austin *Dec. 2024*
Thesis: *Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE*
Advisor: Dr. Keith Hawkins

B.S. in Astrophysics, The University of Florida *May 2021*
summa cum laude
Thesis: *J0010-1735−A Stellar Journey from an R-process Enhanced Ultra Faint Dwarf Galaxy to the Milky Way Halo*
Advisor: Dr. Rana Ezzeddine

APPOINTMENTS

Graduate Student Researcher *2022 - Present*
Department of Astronomy, The University of Texas at Austin
Advisor: Dr. Keith Hawkins

Pre-Doctoral Fellow *Jan. 2026 - May 2026*
Center for Computational Astrophysics, Flatiron Institute
Advisors: Dr. Adrian Price-Whelan, Dr. Carrie Filion, Dr. Aida Behmard

Undergraduate Student Researcher *2019 - 2021*
Department of Astronomy, The University of Florida
Dr. Rana Ezzeddine

SELECT HONORS AND AWARDS

Center for Computational Astrophysics Pre-Doctoral Fellowship *2026*
Flatiron Institute, Simons Foundation
David L. Lambert Graduate Fellowship *2024*
Department of Astronomy, The University of Texas at Austin
Senior Thesis Highest Honors *2021*
Department of Astronomy, The University of Florida

PUBLICATIONS

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First Author

2. *A Detailed Chemical Analysis of the Red Giant Orbiting Gaia BH3: From Lithium to Thorium*
Hackshaw Z., Hawkins K., and Manea C. 2025 **ApJ**, DOI 10.3847/1538-4357/ae1617

1. *[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE*
Hackshaw Z., Hawkins K., Filion C., Horta D., Laporte C. F. P., Carr C., Price-Whelan A. M et al 2024 **ApJ**, DOI 10.3847/1538-4357/ad900e

Contributing Author

6. *An Analytic Model for Stellar Metallicity Gradient Residuals in Cold, Phase-Mixed Galactic Disks*
   Modak S., **Hackshaw Z.**, and Hawkins K. 2026, **ApJL**, DOI 10.48550/arXiv.2606.14850

5. *Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from Kinematic Ridges with Gaia*
   Jurado C., Hawkins K., Hunt J. A. s., **Hackshaw Z.**, et al. 2025, **ApJ**, DOI 10.48550/ arXiv.2601.02494

4. *Exploring the History of Stellar Mergers with Chemistry: Examining the Origins of Massive α-enriched Stars Using Carbon Isotope Ratios*
   Maas Z. G., Hawkins K., Gerber J. M., **Hackshaw Z.**, Manea C., et al. 2025, **ApJ**, DOI 10.48550/arXiv.2509.08900

3. *Optical Spectroscopy Reveals Hidden Neutron-capture Elemental Abundance Differences among APOGEE-identified Chemical Doppelgängers*
   Manea C., Ness M., Hawkins K., Zeimann G., Hogg D. W., Filion C., Griffith E., Johnston K., Casey A., **Hackshaw Z.**, et al. 2025, **ApJ**, DOI 10.48550/arXiv.2508.16717

2. *Discovery of a Metal-poor Red Giant Star with the Highest Ultralithium Enhancement*
   Kowkabany J., Ezzeddine R., Charbonnel C., Roederer I. U., Wang E. X., Li Y., **Hackshaw Z.**, et al 2024, **ApJ**, DOI 10.3847/1538-4357/ad6004/ arXiv.2209.02184

1. *Uranium Abundances and Ages of r-process Enhanced Stars with Novel U II Lines*
   Shah S., Ezzeddine R., Ji A., Hansen T., Catelan M., **Hackshaw Z.**, et al 2023, **ApJ**, DOI 10.3847/1538-4357/acb8af

SELECT SCIENTIFIC PRESENTATIONS
Clickable link

Spectroscopy Workshop Retreat 2026, **Co-Lead** — *August 2026*
Yosemite National Park

Cool Stars, Stellar Systems and the Sun 23, **Poster** — *June 2026*
University of Tokyo, ABC, ISAS-JAXA

Pre-Doctoral Symposium, **Talk** — *May 2026*
Center for Computational Astrophysics; Flatiron Institute

Astronomy Seminar, **Talk** — *March 2026*
Columbia Univeristy

s,i,r- Element Nucleosynthesis (sirEN) Conference, **Poster** — *June 2025*
Italian National Institute for Astrophysics

Milky Way Assembly Tale Conference, **Poster** — *June 2024*
Istituto Nazionale Di Astrofisica

SDSS-V Collaboration Meeting, **Contributing Talk** — *Aug. 2023*
Sloan Digital Sky Survey-V

Stellar Properties from NLTE Radiative Transfer, **Contributing Talk** — *May 2023*
IReNA, Kavli Institute for Cosmological Physics, and the University of Chicago

AAS Meeting, **Poster** — *Jan. 2021*
American Astronomical Society

SELECT OBSERVING EXPERIENCE

PI, *Pollution of FGK White Dwarf Companions*, 7 nights — *Oct. 2024*
Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory

PI, *Chemical Analysis of Gaia BH3*, 7 nights — *June 2024*
Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory

PI, Cosmochronometry with Asteroseismic Ages, *7 nights* — *Feb. 2024*
Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory

COMPUTER SKILLS

Languages: Python, Julia, C++
Software: Turbospectrum, BACCHUS, Spectroscopy Made Hard, Git, GitHub, LaTeX

TEACHING & MENTORSHIP

REU McDonald Observatory Trip Lead — *July 2026*
McDonald Observatory, The University of Texas at Austin

Graduate-Undergraduate Mentorship in astronoMY Mentor — *2024 - Present*
The University of Texas at Austin

Research Mentor, Students: Abigail Skelton and Nachiket Yadav — *2024*
The University of Texas at Austin

Teaching Assistant, AST 325K: Stellar Astronomy — *2023*
The University of Texas at Austin

OUTREACH & SCIENCE COMM.
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National Park Ranger Talk, **Organizer** — *August 2026*
Yosemite National Park

AAS Journal Author Series, **Invited Talk** — *June 2026*
American Astronomical Society

Astronomy Club Guest Lecture, **Invited Talk** — *March 2026*
Davidson Academy Online

Fourth Grade Astronomy Lesson, **Invited Talk** — *Feb. 2026*
Eagle Ridge Elementary

Live Deep Sky Tour Stream, **Invited Talk** — *Dec. 2025*
McDonald Observatory

Future in STEM Career Seminar, **Invited Talk** — *Oct. 2025*
Lynn University

Astronomy Day Distance Learning Presentation, **Invited Talk** — *Sept. 2025*
McDonald Observatory

Explore our Solar System Teacher Workshop, **Invited Co-Lead** — *July 2025*
McDonald Observatory

Astronomy on Tap ATX Presentation, **Contributing Talk** — *Feb. 2025*
Astronomy on Tap Austin, TX

UT STEM Girl Day, **Contributing Activity Lead** — *Feb. 2023 - 2025*
The University of Texas at Austin

Astronomy at the Library Presentation, **Contributing Talk** — *Nov. 2024*
Williamson County Astronomy Club

Board of Visitors Summer Meeting, **Invited Talk** — *July 2024*
McDonald Observatory and the University of Texas at Austin

Girl Scout Space Badge Presentation, **Invited Talk** — *April 2022*
Daisy Troop 41119, Austin, TX

LEADERSHIP & EXTERNAL ACTIVITIES

Social Media and Marketing Manager — *2023 - Present*
Astronomy on Tap, Austin, TX Chapter

Prospective Graduate Student Visit Lead — *2023-2025*

Department of Astronomy, The University of Texas

Assistant Director of Mini Marathons *2020 - 2021*

Morale Team for Dance Marathon at the University of Florida

Vice President of Chapter Relations and Standards *2018 - 2019*

Alpha Chi Omega, Gamma Iota Chapter

PROFESSIONAL EXPERIENCE

Engineering Scientist Associate *2021 - 2022*

Applied Research Laboratories, University of Texas at Austin